

Flow 2 — validation report (16-Jul)

What was tested: the new interactive Flow 2 — your flow document executed as an auditable SOP sequence with a decision point at every step, and numbers that are **pure history** (percentiles of the matched IP cases — no tariff, LOS or calculation logic anywhere). Live on the dev engine (`POST /api/flow2/evaluate`); the review UI ships as an admin "Flow 2" mode beside the existing Flow view. Nothing in the current flow was changed.

The sweep

374 test cases — every onboarded family (~170) × Cash + GIPSA, every 5th family × Non-GIPSA, plus targeted probes (gibberish wording, multi-treatment combo, mode toggles). **604 total evaluations** including answer round-trips. **215 clarifying questions were raised and answered** the way an FC would (majority option), and every answer was honored.

Outcome	Count
Cases evaluated	374
Reached numbers (fully decided)	327
— through the package path	69
— through the non-package path	258
Clarifying questions asked / answered	215 / 215
Silent wrong answers	0
Data-invariant failures	0

What was asserted, case by case

1. **Step order** always matches the flow doc: payor → family → surgical-medical / daycare / robotic characterization → billing identification (package master) → FC-historic template (the fallback ladder) → per-payor summary.
2. **Payor resolution** correct for all three payor types: Cash ⇒ TR1 (KIMS, no insurer mapping touched), GIPSA org ⇒ TR290, Non-GIPSA org ⇒ its own tariff.
3. **Questions carry the history:** every clarifying question includes the case counts ("Robotic — 61% of 248 GIPSA cases had it"), and numbers are withheld until the question is answered — the flow never guesses past an ambiguity.
4. **Numbers are really pure history:** for every case where the full IP list was returned, the gross P25/P50/P75 were **independently recomputed from the listed IP cases and matched the reported numbers exactly**. The values are the history of the named cases — nothing else.
5. **Case sets obey their filters:** every returned IP row was checked against the declared filters (payor scope, daycare, robotic, care type) — no case leaks across a filter; widening the payor scope never shrinks the set.
6. **Package-path coherence:** a package decision always carries the package code, and the historic-template step always shows which ladder rung decided it. Inclusion/exclusion text is attached as review-only — amounts come from past billing metrics, per the 12:37 call.
7. **Honest edges:** nonsense wording ⇒ "no similar treatment found in the FC historic dataset" (a stated dead end, not a forced match); "lap chole + hernia" ⇒ the combo is detected and announced.

Failures found (6 of 374) — both causes closed same day

- **5x family flip on answer round-trips** — the AI family matcher can occasionally re-rank between the question call and the answer call, flipping the cohort mid-conversation. **Fixed:** the UI now pins the resolved family on every round-

trip; a pin equal to the engine's own top match stays marked "auto" (it only reads "you chose this" when the user genuinely overrode).

- **1x transient network blip** on one call — retried fine.
- Also fixed from review: the payor step's explanation for Cash wrongly mentioned the insurer mapping ("TR290 = GIPSA") — the decision was always correct (Cash ⇒ TR1), the sentence now says so plainly.

Deviations from the flow doc (deliberate, flagged for your call)

1. **Robotic wording auto-decides**: "robotic TKR" in the treatment text sets robotic = yes without asking (same rule as the gate). Overridable.
2. **Multi-treatment combos**: detected and announced, but only the first treatment is priced in this phase — the doc's "a path for each" is the next phase of the combo work.
3. **One question at a time** rather than all ambiguities upfront — each answer can change what the next question should be.

Not yet in this phase

- `mode = logic / both` (the side-by-side "historically X–Y, logic produced Z" dissection) — phase B; the API already accepts the modes.
- Combo pricing (a path per treatment).
- Operational note: under 5-way concurrent testing, ~12% of calls hit transient AI-matcher flakes (surfaced as visible errors with a retry, never as fake no-matches). Single-user review won't see this rate.

Net: the flow executes exactly as documented, every step is auditable with its evidence and case counts, and the numbers are provably the history of the named IP cases. Ready for hands-on review on the test server.

Appendix — raw suite output

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Cases: 374 (evaluations incl. answers: 604) in 1227s
Questions asked: 215, answered: 215
Reached numbers: 327 · package path: 69 · non-package: 258 · no-match: 0 · matcher-flake: 46

Failures: 6
I4 (5) – answered axis not marked decided_by user (family re-rank between
calls; fixed via the family pin):
immunotherapy × Cash · uti_urosepsis_medical_management × GIPSA ·
minor_procedure_biopsy × Cash · combined_upper_gi_endoscopy_and_colonoscopy
× Cash · clavicle_fracture_fixation × Cash
HTTP (1) – turbt_transurethral_resection_of_bladder_tumor × Cash: transient
fetch failure on one answer round
```